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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Structural Engineering) (CBCS) Degree Examinations

May - June 2024

22CST202 – FINITE ELEMENT METHOD OF ANALYSIS

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Differentiate between Finite element method and Classical methods.	10	L*2	1	1
b) List the various applications of FEM.	10	L2	1	1
2. a) Differentiate between plane stress and plane strain problems. Write the stress- strain relations for the same.	10	L2	1	1
b) Explain the steps involved in finite element analysis briefly.	10	L2	1	1

Unit – II

3. a) Illustrate the numbering of nodes to minimize the bandwidth.	10	L2	2	2
b) Evaluate the following integrals.				
$\int_0^l L_1^2 dx \quad \int_0^l L_1^3 L_2 dx \quad \oint_A L_1^2 L_2^2 L_3^3 dA \quad \oint_V L_1^2 L_2 L_4 dV$	10	L2	2	1
4. a) Define shape function. List the properties of shape function. Illustrate the variation of shape function for any element.	8	L2	2	1
b) Derive the shape function for quadrilateral element using natural coordinate system or Lagrange Interpolation Technique.	12	L2	2	2

Unit – III

5. a) Differentiate between Iso parametric, Sub parametric and Super parametric elements.	10	L2	3	1
b) Determine the Cartesian coordinate of the point P ($\xi = 0.5, \eta = 0.6$) Refer Fig. Q5 (b).				

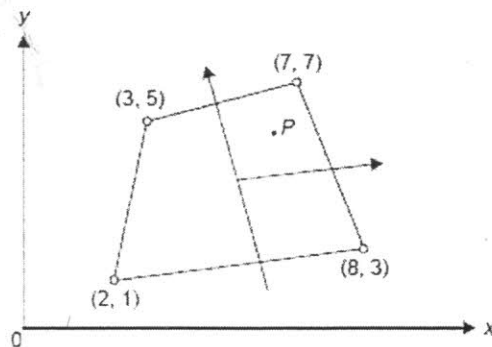


Fig. Q5 (b)

6. Explain the process of developing the Jacobian matrix for CST element.	20	L3	3	3
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Unit – IV

7. Determine the displacements of nodes 1 and 2 in the spring system shown in Fig. Q.7. Use minimum of potential energy principle to assemble equations of equilibrium.

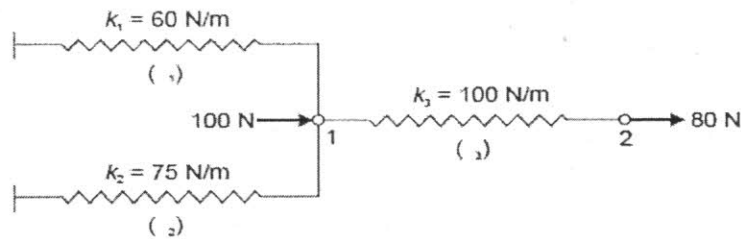


Fig. Q.7

20 L4 4 1,2

8. Determine the deflection under the load for the beam shown in Fig. Q8. Take modulus of elasticity of material as 200 GPa and moment of inertia as $24 \times 10^{-6} \text{ m}^4$.

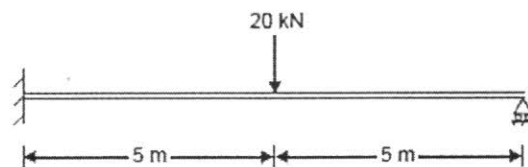


Fig. Q8.

20 L5 4 3

Unit – V

9. Determine the consistent nodal vector due to loads acting on the beam shown in Fig. Q9

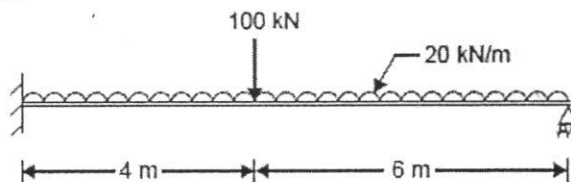


Fig. Q9.

20 L2 5 3

10. a) Explain the finite elements used for shell analysis.
b) List out the assumptions made in bending theory of thin plates. Explain the continuity elements used in classical plate theory.

10 L2 5 1

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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
